

Alex R. Atrio, PhD

Applied AI Project Lead | LLMs and Agentic Systems

Worked with: Mistral AI · European Space Agency · NASA

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 alexratrio.github.io  English, French, Spanish, Catalan, Italian



Summary

- PhD in NLP (EPFL): 6+ years across applied LLM/AI research and delivery
- Leading **€2M ESA project (EVE)**: open-source LLM and agentic platform with **400 users**
- **Full LLM stack**: continued pre-training, instruction fine-tuning, alignment, RAG
- **Agentic systems**: LangGraph + MCP tool registry + A2A multi-agent orchestration
- **Project management**: 2M€ budget, team of 5, stakeholder coordination
- **Prolific scientific publishing**: EACL, EAMT, etc.
- Track record in **grant success**: EuroHPC, Cloud compute, multiple EU/ESA grants
- **Focus**: domain adaptation, low-resource NLP, and regularization methods

Skills

Focus LLMs · Agents · Model reliability · Low-Resource NLP · Machine Translation · Fine-Tuning · RAG
Strategy Scientific Writing · Grant Development · Project Management · Agile
Toolkit PyTorch · Hugging Face · Langchain · Asana · Git · Google Workspace
Coding Python, Prolog, $\LaTeX$

PROFESSIONAL EXPERIENCE

Pi School

Rome, Italy

Project lead – Senior Deep Learning Scientist

2024 - Present

- **Project lead** for **EVE**:
 - * **Managing a 3-year, €2M project** to develop an open-source LLM and agentic platform, grown to **400 users**, for the **European Space Agency**
 - * Weights, code and data **released**. Accepted as **oral in ACL 2026**
 - * **Model**: fine-tuned Mistral Small 3.2 (24B), 5.3B-token domain corpus (continued pre-training, instruction tuning and alignment). RAG (Qdrant, reranking) and hallucination detection
 - * **Orchestration & deployment**: LangGraph workflows with an MCP tool registry and A2A support for multi-agent tool use; serverless (RunPod) inference
 - * **Management responsibilities**: mentoring and guiding 5 engineers, stakeholders discussions, budget oversight.
- Contributor to multiple **winning grant proposals**, including Videoloc (FBK €150k), SatcomLLM (ESA €400k)
- **Project lead** for **Videoloc**: 12-month, €150k on automatic subtitle generation with quality estimation for **FBK**
- **Senior Scientist** for **DVPS**: €29M, 4-year Horizon Europe on multimodal foundation models

HEIG-VD

Lausanne, Switzerland

Teaching Assistant

2019 - 2023

- Final undergraduate NLP courses in French. Guiding students and reviewing laboratory work.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

PhD in Natural Language Processing

2019 - 2023

- Thesis: [Regularization Techniques for Low-Resource Machine Translation](#).
- Supervisor: [Andrei Popescu-Belis](#).
- I also contributed to applied projects with external partners:
 - * Armasuisse - [UNISUB](#): end-to-end MT optimization through adaptive scheduling of training tasks.
 - * Armasuisse - [FamilyMT](#): design systems for closely related source languages.
 - * SNSF Agora - [CRPO](#): development deployment of an interactive poetry generation system.

University Pompeu Fabra & University of Barcelona

Barcelona, Spain

Masters in Cognitive Science

2017 - 2019

- Thesis: [MT Reordering as Preprocessing for Cross-Lingual Sentiment Analysis](#).
- Supervisors: [Toni Badia](#) & [Jeremy Barnes](#).

RESEARCH

I have published across several areas in NLP. For the full list, please visit my [Google Scholar profile](#).

- [Atrio, À.R., et al. \(2026\). EVE: A Domain-Specific LLM Framework for Earth Intelligence \(ACL 26\)](#)
- [Allemann, A., Atrio, À.R., Popescu-Belis, A. \(2025\). Optimizing the training schedule of multilingual NMT using reinforcement learning \(MT Summit 25\)](#)
- [Popescu-Belis, A., Atrio, À.R., et al. \(2023\). GPoeT: a Language Model Trained for Rhyme Generation on Synthetic Data \(EACL 23\)](#)
- [Atrio, À.R., Popescu-Belis, A. \(2022\). On the Interaction of Regularization Factors in Low-Resource NMT \(EAMT 22\)](#)
- [Atrio, À.R., Popescu-Belis, A. \(2021\). Small Batch Sizes Improve Training of Low-Resource NMT \(ICON 21\)](#)
- [Atrio, À.R., Badia, T., Barnes, J. \(2019\). On the Effect of Word Order on Cross-Lingual Sentiment Analysis \(SEPLN 19\).](#)
🏆 Best Paper Award.

Additionally, I have served in the Scientific Committee of the [NASA/ESA workshop on AI FM models and LPS25.](#), and reviewed for ACL, EMNLP, EACL, EAMT, etc.

AWARDS

- [EuroHPC \(2025\)](#): Principal Investigator. Won 3,500 node-hours (**€31,000**) on the MareNostrum5 supercomputer
- [RunPod \(2024\)](#): Principal Investigator. Won Startup & Research Grant: **€3,000** in compute credits
- [SEPLN \(2019\)](#): **Best Paper Award** (first author), on word order in cross-lingual sentiment analysis

INTERESTS

- Projects with state-of-the-art technology.
- Open, friendly and agile working environment.
- Development prospects and responsibility.
- Balance between application and scientific research.

REFERENCES

Upon request.